

# Vibrotactile heatmaps to encourage discussion in a VR environmental city builder

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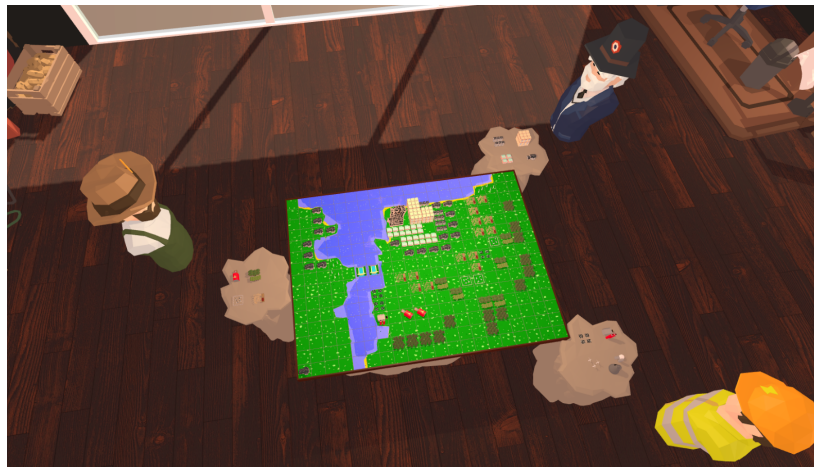


Figure 1: Three users taking part in a decision for urban planning

## ABSTRACT

Decision making can be a difficult process, especially when decision makers do not have access to the full information. While this can be solved through discussion, it is not rare for people to make assumptions based solely on the information they have at hand.

To counter this effect and encourage the discussion in the decision making process, we propose a vibrotactile heatmap, which gives the user a vague representation or intuition of the information they lack access to. Through this, people are encouraged to discuss with other decision makers to reach a consensus.

Furthermore, we implemented this solution in a VR city builder with an environmental sustainability aspect, to raise awareness regarding the problem of divergent interests when it comes to the environment.

**Index Terms:** virtual reality, vibrotactile, heatmap, environment

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## 1 INTRODUCTION

During decision making processes, dialog is of primary importance, as the actors involved in the decision often lack a part of the information needed to reach a consensual decision. This is especially true in situations where no optimal choice exists, as is usually the case regarding environmental sustainability and urban planning. To encourage dialog during decision making, we propose a vibrotactile heatmap providing the user an approximation of the metrics known by the other users. We implemented this system in an application with three users, who must communicate and collaborate to plan the construction of a city while ensuring their own interests.

## 2 HAPTICS HEATMAP

As a means to encourage the discussion between users during decision processes in virtual reality, we propose a vibrotactile heatmap, allowing the user to get insight on the missing information, while keeping the uncertainty coming from the unfamiliarity the attributed role is supposed to have regarding the piece of information transmitted through the vibrotactile heatmap.

While a visual indication would normally make information easier to grasp, it would lead each participant in the decision process to consider that they have all the information they need, and thus would not discuss with the others to try reaching a consensus. By providing a blurry information through the vibrotactile feedback, and by letting the user know that the others do have the missing information, the gap between the user's knowledge and the others' knowledge is made more obvious, which strengthens the *Situation-specific strength* attributed to the other decision makers [1], thus encouraging a discussion oriented behaviour.

We designed a vibrotactile heatmap, inspired by both the visual

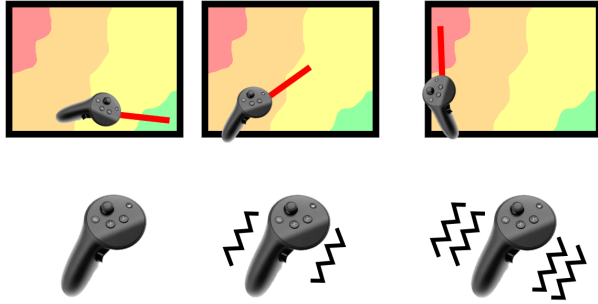


Figure 2: Mapping of the vibration amplitude

| Role               | Metric                                          | Buildings        |
|--------------------|-------------------------------------------------|------------------|
| Farmer             | Pollution<br>(linked to<br>environment quality) | Field            |
|                    |                                                 | Orchard          |
|                    |                                                 | Herd             |
|                    |                                                 | Battery farm     |
| Mayor              | People satisfaction                             | Individual house |
|                    |                                                 | House block      |
|                    |                                                 | Apartment block  |
|                    |                                                 | High rise        |
| Energy<br>producer | Terrain potential                               | Solar panels     |
|                    |                                                 | Windmills        |
|                    |                                                 | Coal             |
|                    |                                                 | Nuclear          |

Table 1: Roles with corresponding metrics and buildings

heatmaps and Geiger counters. While visual heatmap encode a continuous and bounded value as a color (either using saturation, hue or transparency) [2], we propose to use it to encode the importance of a vibration, which is relayed to the user when they move the controller around the heatmap to probe the values. Unlike existing haptic-based data visualisation which rely on mapping color to elevation [3], we choose instead to focus on a real-life metaphor based on a Geiger counter. To make this system compatible with the OpenXR standard<sup>1</sup>, the vibration is sent through the controller.

In order to avoid annoying the user with too much stimuli, the amplitude is set to be null where the value of interest is at the minimum value. On the contrary, a strong vibration is sent to the user's controller where the value is at its possible maximum on the map, as illustrated in Figure 2.

### 3 DEMO

We implemented our system in a VR featuring three users, who must make decisions about urban planning for a town, while having divergent interests. Indeed, it has already been demonstrated that being directly involved through actions makes users more aware and engaged towards environmental issues [4], which is why we decided to make them participate actively in the environment change process.

#### 3.1 Roles

For the sake of the application, we designed 3 characters with specific roles to fulfil. Each character has their own interest and can place some specific buildings. The details for those roles are given in Table 1. Additionally, we implemented three global scoring metrics, known by all users: population, food production, energy production.

<sup>1</sup><https://www.khronos.org/openxr>



Figure 3: The mayor player creating a building

#### 3.2 Setup

At the start of the application, the three users choose their respective roles. They are then placed in front of a map of Saint-Malo (France), with the objective of generating enough housing, energy and food to fulfil the population needs. During the whole process, any user is free to choose to create and place one of their possible buildings, which can be placed anywhere on the map, except for some buildings that can only be built on land.

To choose where to place the building, the user selects a place on the map with a ray coming from their controller. At the chosen position on the map, the building will be shown as a preview of the construction. Furthermore, two pieces of information are given to the user as a means to choose an optimal placement, as illustrated in Figure 3

First, a visual heatmap is displayed to convey information about the metric the user is responsible for. For instance, the mayor in Figure 3 sees the population satisfaction as smiling faces. The second piece of information conveys the missing information for the user. Since they lack the two other metrics, a means of the two other metrics is computed and transmitted in the form of a vibrotactile feedback, as detailed in section 2.

In this case, we compute the mean of the loss caused on the two unknown metrics by the placement of a new building. A loss of 100% on the two metrics would then result in a maximal amplitude, while an absence of change would not trigger any vibration.

Although we chose to map the metrics on the amplitude of the vibration, it would also be possible to change either the duration or the frequency for instance, which would lead the users to approach the information differently.

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